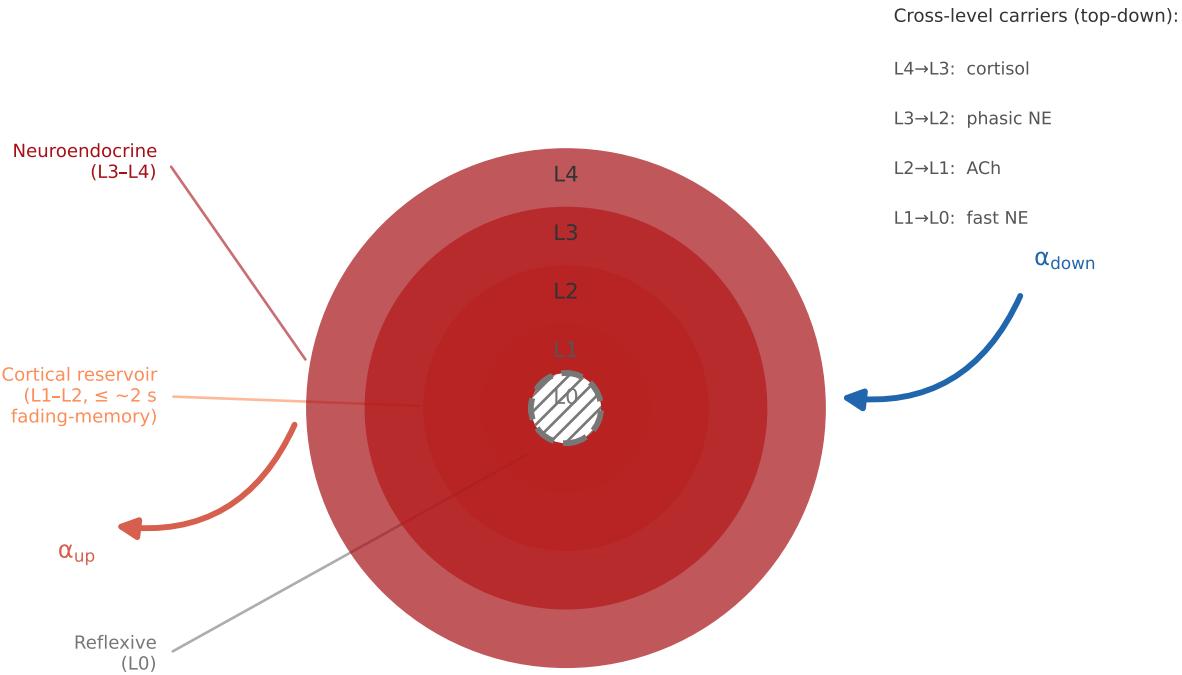


Russian Doll concentric-ring hierarchy



L0 is sub-APGI (no $\Pi_0/\theta_0/\tau_{ign,0}$); supplies sensory drive to L1

Level	τ_{int}	Substrate	Neuromod.	APGI params
L4	$10^2\text{--}10^4$ s (min-hours)	Neuroendocrine (HPA axis)	—	$\Pi_4, \theta_4, \tau_{ign,4}$
L3	$10\text{--}10^2$ s (s-min)	Association cortex (limbic, parietal)	cortisol	$\Pi_3, \theta_3, \tau_{ign,3}$
L2	$0.3\text{--}2$ s (~ 2 s)	Cortical reservoirs (~ 2 s memory)	phasic NE	$\Pi_2, \theta_2, \tau_{ign,2}$
L1	$50\text{--}300$ ms	Local cortical microcircuits	ACh	$\Pi_1, \theta_1, \tau_{ign,1}$
L0 (sub)	$1\text{--}20$ ms	Brainstem, spinal reflex	fast NE	no $\Pi/\theta/\tau_{ign}$